

ORIGINAL RESEARCH

Comparison of Sirolimus- vs Paclitaxel-Coated Balloon Angioplasty for Femoropopliteal Artery Disease



The SIRONA Randomized Noninferiority Trial

Ulf Teichgräber, MD,^a Maja Ingwersen, DVM,^a Thomas Lehmann, PhD,^b Stephanie Platzer, PhD,^b Ina Kunstmann, MD,^a Sabine Steiner, MD,^c Andrej Schmidt, MD,^d Tim Wittig, MD,^d Thomas Zeller, MD,^e Marcus Thieme, MD,^f Hans Krankenberg, MD,^f Martin Werner, MD,^g Marianne Brodmann, MD,^h Nasser Malyar, MD,ⁱ Stefan Müller-Hülsbeck, MD,^j Luca de Paoli, MD,^k Oliver Schlager, MD,^c Martin Austermann, MD,^l Gunnar Tepe, MD,^m Christian Erbel, MD,ⁿ Wulf Ito, MD,^o Rainer Waßmer, MD,^p Lars Maiwald, MD,^q René Aschenbach, MD,^a Christian Wissgott, MD,^r Martin Andrassy, MD,^s Florian Wolf, MD,^t Ulrich Beschorner, MD,^u Maciej Pech, MD,^v Konstantinos Katsanos, MD,^w Dierk Scheinert, MD,^d the SIRONA Study Group*

ABSTRACT

BACKGROUND Paclitaxel-coated balloon (PCB) angioplasty is effective for femoropopliteal interventions. However, the increased mortality risk associated with PCB has not yet been cleared, and alternative antiproliferative drugs may offer additional advantages.

OBJECTIVES The authors aimed to investigate whether sirolimus-coated balloon (SCB) angioplasty achieves noninferior primary vessel patency and clinical efficacy and safety compared with PCB angioplasty.

METHODS SIRONA is a multicenter randomized, controlled, noninferiority trial comparing SCB with PCB angioplasty in patients with symptomatic femoropopliteal artery disease. The study was conducted at 25 centers in Germany and Austria. Patients aged 18 years or older with a single target lesion were randomized 1:1 using permuted blocks, stratified by center. Participants and outcome assessors were masked to allocation. Primary endpoints were primary patency and the composite clinical endpoint of freedom from clinically driven target vessel revascularization, major target limb amputation, and device- or procedure-related death at 12 months based on available case intention-to-treat analysis. Prespecified noninferiority margins were 10%. Long-term follow-up is ongoing.

RESULTS Between April 21, 2021, and September 23, 2022, 482 patients were assigned to SCB (n = 238) or PCB (n = 244) groups. The mean age was 68.0 ± 8.9 years; 311 participants (64.5%) were male; and 464 participants (96.3%) presented with intermittent claudication. Mean lesion length was 8.4 ± 6.1 cm and 160 lesions (33.2%) were chronic total occlusions. Primary patency was achieved in 150 of 203 participants (73.9%) with SCBs and in 149 of 199 participants (74.9%) with PCBs (risk difference: -1.0% ; 95% CI: -9.6% to 7.6% ; $P_{\text{noninferiority}} = 0.019$). However, sensitivity analyses revealed that noninferiority was achieved but statistically not robust. The primary clinical endpoint was achieved in 199 of 220 participants (90.5%) with SCBs and in 202 of 218 participants (92.7%) with PCBs (risk difference: -2.2% (95% CI: -7.7% to 3.2%); $P_{\text{noninferiority}} = 0.003$). These results were confirmed by sensitivity analyses.

CONCLUSIONS Among patients with femoropopliteal artery disease, SCBs demonstrated noninferior clinical important outcomes and comparable improvement in vascular quality of life compared with PCBs. Although the noninferiority of primary patency was not statistically robust, this suggests that SCBs may be a valuable alternative to PCBs. (Sirolimus- vs Paclitaxel-Drug Coated Balloons in Patients With Peripheral Artery Disease [SIRONA]; [NCT04475783](https://doi.org/10.1016/j.jacc.2025.12.017); Head-to-Head Comparison of Sirolimus vs Paclitaxel Drug-Eluting Balloon Angioplasty in the Femoropopliteal Artery [SIRONA]; [DRKS00022452](https://doi.org/10.1016/j.jacc.2025.12.017)) (JACC. 2026;88:780-796) © 2026 Published by Elsevier on behalf of the American College of Cardiology Foundation.

Lower extremity arterial disease is a common condition with a prevalence of 3% to 10% in adults, increasing to up to 20% with age.¹ Femoropopliteal involvement is a characteristic finding in a large proportion of patients with intermittent claudication. Revascularization may be considered in patients with disabling symptoms that persist despite optimal noninvasive therapy, taking into account the procedural risks and anticipated long-term patency. In a subset of patients, claudication may progress to chronic limb-threatening ischemia, a condition associated with substantial morbidity and mortality.^{1,2} According to the 2024 European Society of Cardiology guideline, drug-eluting treatment should be considered as the first-choice strategy for endovascular interventions (Class 2a, Level of Evidence: A).² Society for Cardiovascular Angiography and Interventions consensus guidelines strongly recommend drug-coated balloons (DCBs) as intended definitive therapy for femoropopliteal lesions.³ Postinterventional continuation of pharmacotherapy, exercise, and risk factor control is mandatory.

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Unlike balloon angioplasty, stents carry a risk of in-stent restenosis and may interfere with subsequent bypass graft surgery. In addition, stents are subject to high levels of motion, particularly in the femoropopliteal segment, which can compromise their structural integrity.⁴ All of this together may have contributed to the increased use of paclitaxel-coated balloons (PCBs), which have proven

efficacy.⁵ Subsequent research initially refuted the suspicion⁶ that paclitaxel-coated devices are associated with an increased risk of late mortality. Consequently, the U.S. Food and Drug Administration recently stated that an increased risk of mortality is unlikely,^{7–9} though they recommended that patients receiving paclitaxel-coated devices continue to be monitored. Notably, the recently published SWEDEPAD (Swedish Drug-elution Trial in Peripheral Arterial Disease)-2 trial on patients with intermittent claudication revealed an increased risk of mortality risk at 5 years with paclitaxel-coated devices compared to uncoated balloons or stents. Furthermore, after 1 year, there was no difference in vascular quality of life between groups.¹⁰ In addition, research on coronary arteries showed that sirolimus has stronger immunosuppressive effects and inhibits smooth muscle migration more effectively than paclitaxel.^{11,12} But presumably due to the poorer tissue absorption and retention of sirolimus, these effects were so far only observed when it was continuously released from drug-eluting stents.¹³ The coronary angiographic outcomes of sirolimus-coated balloons (SCBs), including late luminal loss, binary restenosis, and the probability of late luminal enlargement, were worse than those of PCBs.¹¹ These differences may depend on the technological and pharmacologic properties of the SCB type. Notwithstanding, the therapeutic window of sirolimus is wider than that of

**ABBREVIATIONS
AND ACRONYMS**

CEC	= Clinical Events Committee
DCB	= drug-coated balloon
DSMB	= Data Safety Monitoring Board
DUS	= duplex ultrasound
ITT	= intention-to-treat
mITT	= modified intention-to-treat
PCB	= paclitaxel-coated balloon
RCT	= randomized controlled trial
RD	= risk difference
SCB	= sirolimus-coated balloon
TLR	= target lesion revascularization
TVR	= target vessel revascularization
VascuQoL	= Vascular Quality of Life [questionnaire]

From the ^aDepartment of Radiology, Jena University Hospital, Friedrich-Schiller University Jena, Jena, Germany; ^bCenter for Clinical Studies, Jena University Hospital, Friedrich-Schiller University Jena, Jena, Germany; ^cDepartment of Angiology, University Hospital for Internal Medicine II, Medical University Vienna, Vienna, Austria; ^dDepartment of Angiology, University of Leipzig Medical Center, Leipzig, Germany; ^eDepartment of Angiology, University Heart Center Bad Krozingen, University Hospital Freiburg, Bad Krozingen, Germany; ^fDepartment of Vascular Medicine, Sana-Kliniken Oberfranken Coburg GmbH, Coburg, Germany; ^gDepartment of Angiology, Hanusch Hospital Vienna, Vienna, Austria; ^hDepartment of Angiology, Medical University Graz, Graz, Austria; ⁱDepartment of Cardiovascular Medicine, University of Münster, University Hospital Münster, Münster, Germany; ^jDepartment of Radiology, Diakonissenkrankenhaus Hospital Flensburg, Flensburg, Germany; ^kDepartment of Radiology, Landeskrankenanstalten-Betriebsgesellschaft Kärnten Klinikum Klagenfurt am Wörthersee, Klagenfurt, Austria; ^lDepartment of Vascular Surgery, St Franziskus Hospital Münster, Münster, Germany; ^mDepartment of Radiology, RoMed Klinikum Rosenheim, Rosenheim, Germany; ⁿDepartment of Cardiology and Angiology, University Hospital Heidelberg, Heidelberg, Germany; ^oDepartment of Angiology, Cardiovascular Center Klinikverbund Allgäu, Immenstadt, Germany; ^pDepartment of Angiology, Elblandklinikum Radebeul, Radebeul, Germany; ^qDepartment of Angiology, Kreiskrankenhaus Torgau, Torgau, Germany; ^rDepartment of Radiology, Schön Clinic Rendsburg, Rendsburg, Germany; ^sDepartment of Cardiology, Fürst-Stirum-Klinik Bruchsal, Bruchsal, Germany; ^tDepartment of Radiology, University Hospital for Radiology, Medical University Vienna, Vienna, Austria; ^uCoreLab Black Forest, Bad Krozingen, Germany; ^vDepartment of Radiology and Nuclear Medicine, University Hospital Magdeburg, Otto von Guericke University, Magdeburg, Germany; and the ^wDepartment of Radiology, Patras University Hospital, Patras, Greece. *The SIRONA Study Group members are listed in the acknowledgments.

The authors attest they are in compliance with human studies committees and animal welfare regulations of the authors' institutions and Food and Drug Administration guidelines, including patient consent where appropriate. For more information, visit the [Author Center](#).

paclitaxel, reducing the risk of local toxicity.^{14,15} Sirolimus has a considerably less lipophilic profile than paclitaxel, which requires more complex processing for balloon coating to ensure both sufficient adhesion to the balloon and bioavailability. Recently, sirolimus has been used successfully as an active ingredient on DCBs that are already on the market. Results from 3 prospective single-arm studies on 2 SCB types with different coating technologies suggest strong long-term efficacy.¹⁶⁻¹⁸

The SIRONA (Sirolimus- vs Paclitaxel-Drug Coated Balloons in Patients With Peripheral Artery Disease) trial aimed to investigate whether femoropopliteal SCB angioplasty is noninferior to PCB angioplasty in preventing restenosis, as measured by primary patency, and in achieving the clinically relevant outcome of freedom from clinically driven target vessel revascularization (TVR), major target limb amputation, and device- or procedure-related death.

METHODS

STUDY DESIGN AND PARTICIPANTS. The SIRONA trial is a multicenter, randomized controlled, non-inferiority trial, initiated by the investigators to compare the efficacy and safety of SCB angioplasty with PCB angioplasty in patients with symptomatic femoropopliteal artery disease. Recruitment was carried out by 25 centers in Germany and Austria. The trial design has been described previously.¹⁹

In brief, patients aged 18 years or older with clinical symptoms of Rutherford categories 2 to 4 were eligible. Single de novo or restenotic lesions measuring at least 2 cm and up to 20 cm in length with $\geq 70\%$ visual diameter stenosis should be located within the superficial femoral artery and/or the proximal segment of the popliteal artery. Target lesions should have been successfully predilated. Patency ($< 50\%$ stenosis) of inflow and outflow vessels, including the middle and distal popliteal arteries and at least 1 infrapopliteal artery, had to be ensured. Main exclusion criteria were severe calcification, in-stent restenosis, subintimal or failed guidewire crossing of the target lesion, and flow-limiting dissection of the target lesion after predilation. The full list of eligibility criteria can be found in the Clinical Investigation Plan ([Supplemental material](#)).

All patients provided written informed consent to any procedures specific to the study. Sex was reported by the investigator as either male or female. Study data were monitored by a contract research organization (VascuScience GmbH). A blinded core laboratory (CoreLab Black Forest GmbH) evaluated duplex ultrasound and angiographic images for

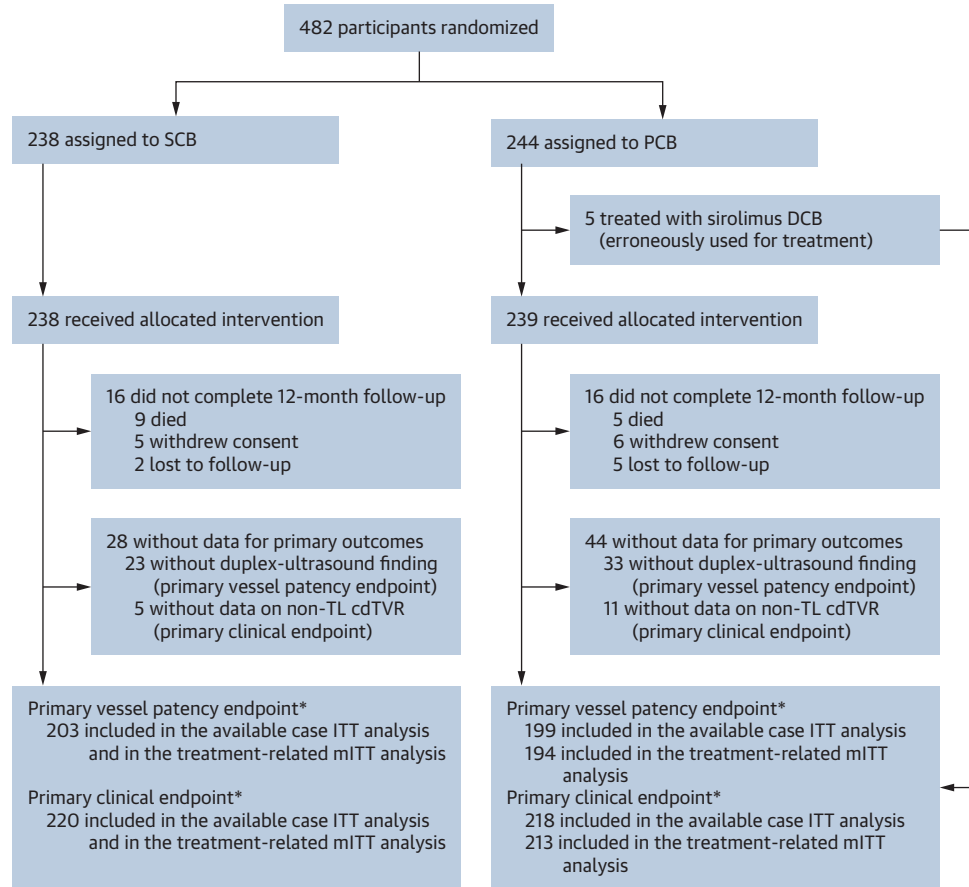
lesion characteristics, revascularization procedures, and patency. An external, masked Clinical Events Committee (CEC) adjudicated the primary outcome measures, and an external Data Safety Monitoring Board (DSMB) reviewed and evaluated the adverse events, using all available information including site case report forms, clinical reports, and duplex ultrasound as well as angiographic images.

The trial is sponsored by the Friedrich-Schiller University Jena and is coordinated by the Centre of Clinical Studies (Friedrich-Schiller University Jena). It is conducted in accordance with the principles of the Declaration of Helsinki, ISO14155 (clinical investigations of medical devices for human subjects), and applicable regulatory requirements. The protocol has been approved by an independent Ethics Committee (Ethics Committee of the Friedrich-Schiller University Jena; reference number 2020-2012-MPG_ff) and by the local Ethics Committees of each participating site. The study is ongoing, with further scheduled follow-ups at 2, 3, 4, and 5 years. SIRONA has been prospectively registered with ClinicalTrials.gov ([NCT04475783](#)) and the German Clinical Trials Register ([DRKS00022452](#)).

RANDOMIZATION AND MASKING. After confirmation of eligibility, study participants were randomized (1:1) to receive either SCB or PCB angioplasty. The randomization process was overseen by an independent statistician (Center for Clinical Studies, Friedrich-Schiller University Jena, University Hospital Jena) who used a computerized random number generator (nQuery Advisor, version 7.0, Statistical Solutions, Ltd) to generate varying block sizes, stratified by center. Allocation concealment until intervention was ensured using numbered, sealed, opaque envelopes. After successful predilation (ie, no flow-limiting dissection), investigators randomly assigned participants to their designated treatment group within each envelope. Trial participants and outcome assessors, including core laboratory staff members, and the CEC were masked to the treatment allocation, whereas the investigators, the treating physicians, and the DSMB were not.

PROCEDURES. The study procedures have been previously published in detail.¹⁹ Percutaneous transluminal angioplasty of lower extremity arteries should have been performed according to the current guidelines² and best practice. After angiographic assessment, predilation of the target lesion for at least 60 seconds after intraluminal guidewire crossing was mandatory, but prolonged predilation for 180 seconds was strongly recommended. The nominal balloon diameter should be 1 mm less than

FIGURE 1 Study Profile



The primary vessel patency endpoint was primary patency. The primary clinical endpoint was freedom from the composite of clinically driven target vessel revascularization (cdTVR), major target limb amputation, and device- or procedure-related death. The available case intention-to-treat (ITT) analysis included all participants who were assigned to each treatment group and whose outcome was known. The treatment-related modified intention-to-treat (mITT) analysis included only participants who received their assigned treatment and had a known outcome. *The number of participants analyzed for the primary endpoints cannot be derived directly from subtraction of those with missing data, as specified in the text, because the primary endpoints could be analyzed in participants with known target lesion (TL)/vessel revascularization despite failure to complete 12 months of follow-up or missing duplex ultrasound outcomes. DCB = drug-coated balloon; PCB = paclitaxel-coated balloon; SCB = sirolimus-coated balloon.

the distal reference vessel diameter. Patients were considered ineligible if a flow-limiting dissection occurred during predilation. Predilation was followed by randomization. The index procedure was then performed with DCB angioplasty using either SCB (intervention group) or PCB (control group). The nominal diameter of the DCB should have been matched to the distal reference vessel diameter and the length should extend approximately 1 cm beyond each end of the target lesion. DCB inflation should be maintained for at least 60 seconds, but prolonged inflation for 180 seconds was strongly recommended. If more than 1 DCB was required to cover the lesion,

the DCBs should overlap for at least 1 cm. It was essential that all devices used in the study had a European Certificate of Conformity (CE mark), were commercially available, and were used according to the manufacturer's instructions. No other adjunctive treatment was allowed, including the use of re-entry devices, atherectomy devices, cutting or scoring balloons, drug-eluting stents, or noninvestigational DCBs.

Participants in the intervention group received the Magic Touch PTA SCB catheter (Concept Medical Inc). The polymer-free balloon coating consisted of sub-micron-sized particles of phospholipid encapsulated

TABLE 1 Participant Characteristics at Randomization by Treatment Group

	SCB Group (n = 238)	PCB Group (n = 244)
Age, y	67.7 ± 9.1	68.3 ± 8.6
Sex		
Male	152 (63.9)	159 (65.2)
Female	86 (36.1)	85 (34.8)
Body mass index, kg/m ²	27.0 ± 4.4	26.9 ± 4.5
Smoking status		
Current smoker	102/236 (43.2)	114/241 (47.3)
Former smoker	97/236 (41.1)	94/241 (39.8)
Never smoker	37/236 (15.7)	33/241 (13.7)
Comorbidities		
Diabetes mellitus	81 (34.0)	81 (33.2)
Hypertension	209 (87.8)	209 (85.7)
Hyperlipidemia	181 (76.1)	195 (79.9)
Coronary artery disease	71 (29.8)	85 (34.8)
Congestive heart failure	16 (6.7)	10 (4.1)
Cardiac arrhythmias	28 (11.8)	25 (10.2)
Renal insufficiency	41 (17.2)	45 (18.4)
Pulmonary disease	43 (18.1)	46 (18.9)
Medical history		
Myocardial infarction	37 (15.5)	43 (17.6)
Cerebrovascular disease	23 (9.7)	20 (8.2)
Rutherford classification		
Category 2	58 (24.4)	65 (27.0)
Category 3	173 (72.7)	168 (68.9)
Category 4	5 (2.1)	10 (4.1)
Category 5	2 (0.8)	0 (0)
Ankle-brachial index of target limb ^a	0.65 ± 0.21	0.67 ± 0.24

Values are mean ± SD, n (%), or n/N (%). There were no significant differences between the groups. ^aData for the ankle-brachial index of the target limb are shown for 227 and 235 participants of the SCB and PCB groups, respectively.
PCB = paclitaxel-coated balloon; SCB = sirolimus-coated balloon.

sirolimus at a concentration of 1.27 µg/mm². The particles, called nanocarriers, diffuse into the vessel wall, where sirolimus is released when the pH changes.²⁰ In the control group, only PCBs with proven efficacy in the treatment of peripheral artery disease were allowed. Efficacy over 2 years had to be demonstrated in randomized controlled trials (RCTs).

Postdilation of lesions with residual diameter stenosis ≥30% and <50% by visual estimation after DCB angioplasty could be performed with the used DCB or a standard uncoated balloon at the discretion of the investigator. In case of residual diameter stenosis ≥50% or trans-lesion gradient >10 mm Hg, postdilation should be performed. Nevertheless, should the residual diameter stenosis persist at ≥50% or the trans-lesion gradient remain at >10 mm Hg, bailout stenting was mandatory. Only bare nitinol stents were permitted.

Prior to the procedure, participants were required to receive a prescription for antiplatelet therapy according to current guidelines² and the center's standard of care. Dual antiplatelet therapy was recommended for at least 3 days before the procedure or as a loading dose at the procedure. Participants must receive appropriate anticoagulation during the index procedure. After the procedure, dual antiplatelet therapy was strongly recommended for at least 4 weeks, after which single antiplatelet therapy should be continued indefinitely.

OUTCOMES. The primary vessel patency endpoint was primary patency based on available case intention-to-treat (ITT) analysis²¹ at 12 months (356 ± 45 days after randomization). Primary patency was defined as the absence of target lesion restenosis as assessed by duplex ultrasound (DUS) and indicated by a peak systolic velocity ratio of >2.4 or total occlusion, as adjudicated by the masked core laboratory, without the need for additional target lesion revascularization (TLR). The CEC adjudicated all primary vessel patency outcome measures that had been assessed as borderline by the core laboratory.

The primary clinical endpoint was a composite of freedom from clinically driven TVR, major target limb amputation, and device- or procedure-related death based on available case ITT analysis at 12 months (365 ± 45 days after randomization). TVR included TLR and non-TLR TVR. It was considered clinically driven in participants with symptoms and a decrease in the ankle-brachial index of ≥20% or >0.15 from postprocedure or a peak systolic velocity ratio >2.4 as assessed by DUS. Major amputation was defined as amputation above the ankle.

Prespecified secondary endpoints were the primary vessel patency and clinical endpoints in the treatment-related modified intention-to-treat (mITT) population at 12 months. Kaplan-Meier estimates of primary and secondary patency, freedom from clinically driven TLR and TVR, and amputation-free survival at 180 and 365 days were analyzed in the ITT and the treatment-related mITT populations. Secondary patency was defined as patency obtained with or without the use of an additional endovascular procedure. Death from any cause was reported with an indication of the cause of death. Other prespecified secondary endpoints were functional outcomes at 6 and 12 months, including clinical improvement by at least 1 Rutherford category, hemodynamic improvement defined as an increase in ankle-brachial index by 0.15 or to at least 0.9, change in pain-free and maximum walking distance, and

change in the vascular and health-related quality of life. To assess the walking distance, the 6-minute walk test and participant self-assessment were mandatory,²² whereas the treadmill test was optional. Participants' vascular quality of life was expressed as a score derived from the 25-item Vascular Quality of Life (VascuQoL) questionnaire,²³ and their health-related quality of life was expressed as an index derived from the European Quality of Life 5 Dimensions 3 Level instrument.²⁴ Functional outcomes were assessed using the available case ITT analysis excluding participants who underwent TVR prior to the follow-up. The treatment interaction of lesion length was analyzed in a prespecified subgroup analysis.

STATISTICAL ANALYSIS. Based on the results of the randomized COMPARE (Compare I Pilot Study for the Treatment of Subjects With Symptomatic Femoropopliteal Artery Disease) study,²⁵ which showed a primary patency rate of 89% at 12 months after PCB, and based on the IN.PACT global study (NCT01609296),²⁶ which showed a freedom from the composite safety endpoint of 94% after PCB, a sample size of 430 patients for the primary vessel patency endpoint and 280 patients for the primary clinical endpoint was required to detect noninferiority with a power of 89% for each endpoint and 80% for both endpoints. Assuming a drop-out rate of 10%, the minimum number of patients required was 478. Noninferiority was tested using the 1-sided Farrington-Manning test at a significance level of 0.05, which was set at 0.025 as determined by the Bonferroni correction to account for the 2 primary endpoints. Each of the 2 primary noninferiority hypotheses was assessed by determining whether the lower bound of the 95% CI of the difference in the proportion of events would exclude a difference in favor of the PCB group of 10%. The 10% noninferiority margin was justified by the known magnitude of vessel patency with PCB compared to standard balloon angioplasty of 45% (12-month primary patency risk ratio: 1.45; 95% CI: 1.27-1.66), as demonstrated in a meta-analysis.⁵ The margin was also justified for the primary clinical endpoint, which was expected to be driven primarily by TVR, because PCB had previously been shown to reduce the absolute risk of 12-month TVR by 17% compared to standard balloon.²⁷ The margins of 10% were considered to be achievable within real-world feasibility. In addition, risk ratios and odds ratios are reported for the primary and the binary secondary endpoints.

Primary endpoint analyses were performed using an available case ITT analysis, which included all participants assigned to each treatment group whose outcome was known. Prespecified sensitivity analyses were performed on the treatment-related mITT population, which only included participants, who received their assigned treatment and had a known outcome. To account for missing data, further sensitivity analyses of the primary endpoints were performed by multiple imputation using the fully conditional specification approach under the missing-at-random assumption. The imputation model included baseline participant and lesion characteristics.

Secondary categorical endpoints were assessed using the chi-square test or Fisher exact test. Effect estimates are expressed as risk differences with 95% CIs. Secondary continuous endpoints were assessed using the 2-sided independent samples Student's *t*-test or the Mann-Whitney *U*-test. Effect estimates are presented as mean $\pm$ SD differences. The ordinal endpoint of change in Rutherford category was assessed by ordinal regression. The effect estimate is presented as an OR. Kaplan-Meier estimates with HRs and log-rank *P* values are reported for primary and secondary patency, freedom from TLR, freedom from TVR, and amputation-free survival. Categorical variables are presented as frequencies and percentages, and continuous variables as mean $\pm$ SD or median (Q1-Q3). Prespecified subgroup analyses of the primary endpoints were performed for lesion length ≥ 15 cm vs < 15 cm by fitting binary logistic regression models and using the OR as measure of effect. For the primary noninferiority endpoints, 1-tailed *P* values < 0.025 , and for all secondary endpoints, 2-tailed *P* values < 0.05 were considered significant. Statistical analyses were performed using SAS software version 9.4 (SAS Institute Inc).

RESULTS

A total of 482 patients, enrolled between April 21, 2021, and September 23, 2022, were randomized to receive SCB angioplasty (*n* = 238) or PCB angioplasty (*n* = 244). At 12 months, 402 participants (83.4%) had completed the follow-up for the primary efficacy endpoint and 438 (90.9%) for the primary safety endpoint (Figure 1).

Baseline participant, lesion, and procedure characteristics were similar between groups (Tables 1 and 2). The mean age was 68.0 ± 8.9 years, 171 participants (35.5%) were female and 311 (64.5%) were male. A total of 464 participants (96.3%) had intermittent

TABLE 2 Lesion Baseline and Procedural Characteristics

	SCB Group (n = 238)	PCB Group (n = 244)	Difference ^a (95% CI)
Lesion characteristics reported by the core laboratory			
Lesion length, mm	84.3 ± 61.7	84.1 ± 60.1	0.2 (–11.0 to 11.3)
Long lesions, ≥ 15 cm	34/235 (14.5)	41/234 (17.5)	–3.1% (–9.7% to 3.6%)
Total occlusion	81 (34.0)	79 (32.4)	1.7% (–6.8% to 10.1%)
Length of occluded segment, mm	50.2 ± 44.3	54.1 ± 44.9	–3.9 (–17.9 to 10.1)
Calcification of the target lesion			OR: 1.0 (0.7-1.4)
PACSS grade			
0	18/223 (8.1)	14/219 (6.4)	
1, unilateral, <5 cm	40/223 (17.9)	42/219 (19.2)	
2, unilateral, ≥5 cm	1/223 (0.4)	0/219 (0)	
3, bilateral, <5 cm	100/223 (44.8)	102/219 (46.6)	
4, bilateral, ≥5 cm	64/223 (28.7)	61/219 (27.9)	
Reference vessel diameter, mm	5.0 ± 0.7	5.1 ± 0.7	–0.02 (–0.2 to 0.2)
Stenosis before treatment, % of vessel diameter	83.0 ± 16.1	81.7 ± 17.5	1.3 (–1.7 to 4.4)
Patent run-off vessels before the procedure, n ^b			–0.1 (–0.3 to 0.2)
0	30/178 (16.9)	25/187 (13.4)	
1	46/178 (25.8)	49/187 (26.2)	
2	58/178 (32.6)	67/187 (35.8)	
3	44/178 (24.7)	46/187 (24.6)	
Lesion characteristics reported by the study sites			
Location of the target lesion			
Superficial femoral artery, proximal	35 (14.7)	46 (18.9)	–4.1 (–10.9 to 2.6)
Superficial femoral artery, mid segment	115 (48.3)	114 (46.7)	1.6 (–7.4 to 10.6)
Superficial femoral artery, distal	135 (56.7)	143 (58.6)	–1.9 (–10.8 to 7.0)
Popliteal artery, proximal	36 (15.1)	50 (20.5)	–5.4 (–12.2 to 1.5)
Popliteal artery, mid segment	1 (0.4)	2 (0.8)	–0.04 (–1.8 to 1.0)
Calcification of the target lesion			OR: 1.1 (0.8-1.6)
None or mild	114 (47.9)	123 (50.4)	
Moderate	117 (49.2)	116 (47.5)	
Severe	7 (2.9)	5 (2.0)	
Patent run-off vessels before the procedure, n			–0.1 (–0.2 to 0.1)
1	46 (19.3)	39 (16.0)	
2	84 (35.3)	87 (35.7)	
3	108 (45.4)	118 (48.4)	

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claudication of Rutherford category 2 or 3. The mean ankle-brachial index was 0.66 ± 0.23 . The mean lesion length was 8.4 ± 6.1 cm with 160 lesions (33.2%) completely occluded. In addition, 125 of 442 lesions (28.3%) were assessed as calcification grade 4 by the core laboratory using the PACSS (Peripheral Arterial Calcium Scoring System). A total of 86 lesions (17.8%) involved the proximal popliteal artery. All participants assigned to receive SCBs underwent SCB angioplasty. Of the 244 participants assigned to receive PCBs, 239 were treated accordingly and 5 were erroneously treated with SCBs (Table 3). Predilation was performed in all lesions, post-dilation in 158 lesions (32.8%), and bailout stenting in 106 lesions (22.0%). National Heart, Lung, and Blood Institute type D and E dissections occurred in

177 of 450 lesions (39.3%). Mean diameter stenosis improved from $83.0\% \pm 16.1\%$ preprocedure to $24.6\% \pm 11.9\%$ postprocedure in the SCB group and from $81.7\% \pm 17.5\%$ to $27.6\% \pm 11.7\%$ in the PCB group. Antiplatelet medication and statin therapy at the time of discharge and at follow-ups were similar between groups (Supplemental Table 1).

The endpoint of primary patency at 12 months was achieved in 150 of 203 participants (73.9%) in the SCB group and in 149 of 199 participants (74.9%) in the PCB group (risk difference [RD]: –1.0%; 95% CI: –9.6% to 7.6%; $P_{\text{noninferiority}} = 0.019$). The composite primary clinical endpoint occurred in 199 of 220 participants (90.5%) in the SCB group and in 202 of 218 participants (92.7%) in the PCB group (RD: –2.2%; 95% CI: –7.7% to 3.2%;

TABLE 2 Continued

	SCB Group (n = 238)	PCB Group (n = 244)	Difference ^a (95% CI)
Procedural characteristics, core laboratory reported			
Length of treated segment, mm	117.9 ± 62.1	119.3 ± 63.3	−1.4 (−12.8 to 10.0)
Dissection after all treatment, final angiogram			OR: 0.9 (0.6-1.4)
No dissection	40/219 (18.3)	35/231 (15.2)	
NHLBI type			
B	85/219 (38.8)	111/231 (48.1)	
C	1/219 (0.5)	1/231 (0.4)	
D, delayed but complete distal flow	92/219 (42.0)	78/231 (33.8)	
E, delayed antegrade flow	1/219 (0.5)	6/231 (2.6)	
Other procedural complications			
Perforation	1 (0.4)	0/243 (0)	0.4% (−0.5% to 1.3%)
Distal embolization	4 (1.7)	3/243 (1.3)	0.4% (−1.7% to 2.6%)
Arteriovenous fistula	2 (0.8)	2/243 (0.8)	0.002% (−1.7% to 1.7%)
Target lesion thrombosis ^c	1 (0.4)	1/243 (0.4)	0.001% (−1.2% to 1.2%)
Stent fracture	1 (0.4)	0/243 (0)	0.4% (−0.5% to 1.3%)
Bleeding	2 (0.8)	1/243 (0.4)	0.4% (−1.0% to 1.9%)
Residual stenosis (final angiogram), % of vessel diameter	24.7 ± 11.6	27.6 ± 11.7	−2.7 (−4.9 to −0.6)
Residual stenosis >30%	74/236 (31.4)	89/242 (36.8)	−5.4% (−14.0% to 3.1%)
Patent run-off vessels after the procedure, n ^b			−0.02 (−0.3 to 0.2)
0	20/194 (10.3)	19/197 (9.6)	
1	56/194 (28.9)	54/197 (27.4)	
2	65/194 (33.5)	71/197 (36.0)	
3	53/194 (27.3)	53/197 (26.9)	
Procedural characteristics reported by the study sites			
Predilation performed	238 (100)	244 (100)	0%
Predilation length, mm	75.5 ± 40.0	82.5 ± 48.4	−7.0 (−15.0 to 0.9)
Diameter of predilation balloon, mm	4.6 ± 0.7	4.8 ± 3.6	−0.2 (−0.7 to 0.3)
Inflation pressure of the predilation balloons, atm	9.7 ± 2.5	9.9 ± 2.7	−0.2 (−0.8 to 0.3)
Predilation time, min	2.3 ± 0.9	2.2 ± 0.9	0.03 (−0.2 to 0.2)
Residual stenosis after predilation, % of vessel diameter	33.2 ± 17.3	32.0 ± 16.9	1.1 (−2.0 to 4.2)
DCBs, n			−0.01 (−0.1 to 0.9)
1	176 (73.9)	177 (72.5)	
2	56 (23.5)	62 (25.4)	
3	6 (2.5)	5 (2.0)	
Length of DCBs, mm	94.2 ± 41.7	94.6 ± 35.2	−0.4 (−7.3 to 6.6)
Diameter of DCBs, mm	5.4 ± 0.7	5.4 ± 0.7	0.02 (−0.11 to 0.2)
Inflation pressure of the DCBs, atm	8.6 ± 2.3	9.2 ± 3.1	−0.6 (−1.2 to −0.2)
Dilation time of DCB angioplasty, min	2.8 ± 0.7	2.7 ± 0.5	0.09 (−0.03 to 0.2)
Residual stenosis after DCBs, % of vessel diameter	16.8 ± 14.9	18.3 ± 16.6	−1.4 (−4.3 to 1.4)
Adjunctive treatment performed ^d	92 (38.7)	107 (43.9)	−5.2% (−14.2% to 3.6%)
Postdilation with uncoated balloon	75 (31.6)	83 (34.0)	−2.4% (−10.8% to 6.1%)
DCB angioplasty ^e	7 (2.9)	16 (6.6)	−3.6% (−7.4% to 0.2%)
Bailout stent implantation ^f	56 (23.5)	50 (20.5)	3.0% (−4.4% to 10.5%)
Residual stenosis after adjunctive treatment, %	6.7 ± 8.1	10.1 ± 10.3	−3.4 (−6.2 to −0.7)

Values are mean ± SD or n/N (%) unless otherwise indicated. ^aThe comparison is reported as percentage points for differences in percentages, as mean difference for differences in numbers, or as OR for the ordinal characteristics of degree of calcification and the type of dissection. ^bPatency of outflow vessels was defined as <50% diameter stenosis. ^cNo fresh thrombus. ^dIn 133 of the 199 participants (66.8%) who underwent adjunctive treatment, a dissection had been diagnosed by the investigators. ^eAn SCB of another vendor (SELUTION SLR, Med Alliance SA) was used as adjunctive treatment in a participant of the intervention group. ^fIn 58 of the 106 participants (54.7%) who underwent stent implantation, the treatment was performed because of a dissection.

DCB = drug-coated balloon; NHLBI = National Heart, Lung, and Blood Institute; PACSS = Peripheral Arterial Calcium Scoring System; other abbreviations as in [Table 1](#).

$P_{\text{noninferiority}} = 0.003$) ([Central Illustration](#)). Therefore, the primary analyses confirmed the noninferiority of SCBs to PCBs. However, multiple imputation for missing data ensured the robustness of

noninferiority for the primary clinical outcome, but not for primary patency. This finding was consistent with the results of the sensitivity analyses of the primary endpoints in the treatment-related mITT

TABLE 3 DCB Types Used for the Index Procedure in the Control Group

	Drug Dose	Excipient	No. of Balloons (n = 325)	No. of Participants (n = 244)
PCB				
Luminor 35 ^a	3.0 µg paclitaxel/mm ²	Organic ester	106 (32.6)	83 (34.0)
Lutonix ^b	2.0 µg paclitaxel/mm ²	Polysorbate and sorbitol	59 (18.2)	48 (19.7)
Ranger ^c	2.0 µg paclitaxel/mm ²	Acetyltributyl-citrate	49 (15.1)	36 (14.8)
IN.PACT Admiral ^d	3.5 µg paclitaxel/mm ²	Urea	45 (13.8)	35 (14.3)
Stellarex ^e	2.0 µg paclitaxel/mm ²	Polyethylene-glycol	18 (5.5)	13 (5.3)
Orchid ^f	3.0 µg paclitaxel/mm ²	Magnesium- stearate	13 (4.0)	9 (3.7)
SeQuent Please OTW ^g	3.0 µg paclitaxel/mm ²	Resveratrol	11 (3.4)	6 (2.5)
IN.PACT Pacific ^d	3.0 µg paclitaxel/mm ²	Urea	3 (0.9)	1 (0.4)
ELUTAX ^h	2.2 µg paclitaxel/mm ²	Dextran	1 (0.3)	1 (0.4)
Combination ⁱ			9 (2.8)	4 (1.6)
Unknown DCB type			4 (1.2)	3 (1.2)
SCB^k				
Magic Touch ^l	1.7 µg sirolimus/mm ²	Phospholipid	7 (2.2)	5 (2.0)

Values are n (%). ^aVascular. ^bBecton, Dickinson, and Company. ^cBoston Scientific. ^dMedtronic. ^ePhilips. ^fAcotec Scientific. ^gB Braun. ^hAR Baltic Medical. ⁱConcept Medical. ^jFour participants received combinations of different PCBs: IN.Pact Admiral, SeQuent Please, and Stellarex, IN.Pact Admiral and SeQuent Please, Luminor and Orchid, and Luminor and Ranger. ^kFive participants of the control group were erroneously treated with SCB.

Abbreviations as in [Table 3](#).

([Figure 2](#), [Table 4](#)). Post hoc noninferiority tests of the relative effect measures of the primary outcomes are provided in [Supplemental Table 2](#).

Kaplan-Meier estimates for primary and secondary patency at 365 days were not significantly different between groups (SCBs 74.0% vs PCBs 81.9%, log-rank $P = 0.27$; and SCBs 77.0% vs PCBs 84.5%, log-rank $P = 0.38$, respectively) ([Supplemental Table 3](#)). DUS-defined, site-reported binary restenosis at 6 months was significantly higher with SCBs (RD: 13.8%; 95% CI: 6.8%-20.8%; $P < 0.001$) ([Supplemental Table 4](#)).

There were no significant differences in the cumulative incidences of clinically driven TLR, the composite clinical outcome, and the composite of major target limb amputation, and all-cause death at 12 months between the groups (HR: 1.22; $P = 0.58$; HR: 1.40; $P = 0.31$; and HR: 1.69; $P = 0.30$, respectively) ([Figure 3](#)). This was also true for the 12-month Kaplan-Meier estimates of freedom from clinically driven TLR (SCBs 93.1% vs PCBs 95.6%; log-rank $P = 0.58$), freedom from clinically driven TVR, amputation-free survival, and all-cause mortality ([Supplemental Figures 1 to 6](#), [Supplemental Table 5](#)). Causes of death are shown in [Supplemental Table 6](#).

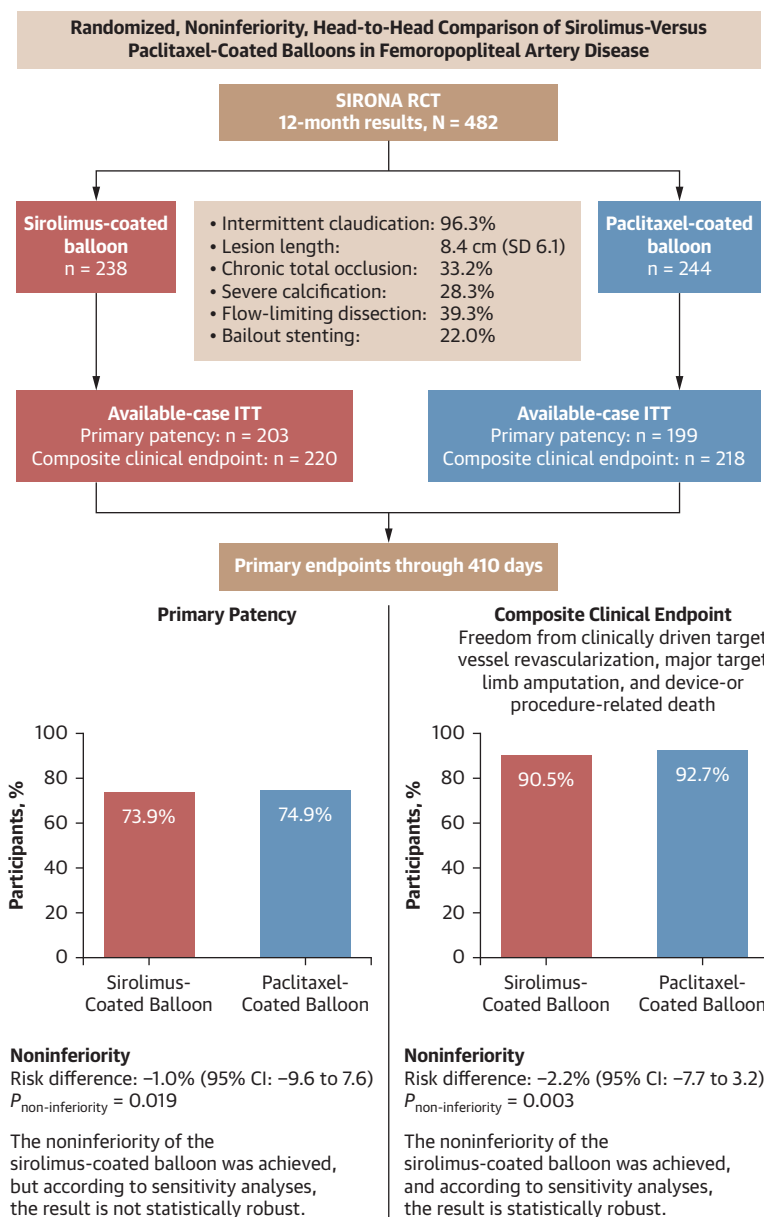
The proportion of participants who improved clinically by at least 1 Rutherford category did not differ significantly between groups (RD: -1.2%; 95% CI: -7.3% to 4.9%; $P = 0.75$) ([Table 4](#), [Supplemental Table 7](#)). However, participants who underwent SCB angioplasty were less likely to show greater clinical improvement than those who

received PCBs (OR: 0.7; 95% CI: 0.4-1.0; $P = 0.043$) ([Table 5](#), [Supplemental Figure 7](#)). There was no significant difference between groups in the proportion of participants who achieved a hemodynamic improvement (RD: -0.3%; 95% CI: -9.6% to 8.9%; $P > 0.99$) ([Table 4](#), [Supplemental Table 7](#)). The same was true for changes in maximum and pain-free walking distance, vascular quality of life, and health-related quality of life ([Table 5](#), [Supplemental Tables 8 and 9](#)). There was no significant difference between the groups in the proportion of participants achieving the minimal important difference for improvement or worsening in the VasculQoL score at 12 months (minimal important difference value for improvement: 0.87, minimal important difference value for worsening: 0.23)²⁸ ([Supplemental Figure 8](#)). In both study arms, the improvement was greatest in the VasculQoL domains of pain and activities ([Supplemental Figure 9](#)).

Serious adverse events according to ISO 14155:2020 were reported in 98 of 238 participants (41.2%) in the SCB group and 89 of 244 participants (36.5%) in the PCB group ($P = 0.30$) ([Supplemental Tables 10 and 11](#)).

The prespecified subgroup analysis showed no significant treatment interaction of lesion length ≥ 15 cm or < 15 cm on primary patency (OR: 0.74; 95% CI: 0.26-2.04; and OR: 0.94; 95% CI: 0.56-1.58, respectively; $P_{\text{interaction}} = 0.68$) and on the primary safety composite (OR: 0.75; 95% CI: 0.21-2.62; and OR: 0.56; 95% CI: 0.23-1.36, respectively; $P_{\text{interaction}} = 0.71$) ([Figure 4](#)).

CENTRAL ILLUSTRATION Key Characteristics of the SIRONA Trial



Teichgräber U, et al. JACC. 2026;88(8):780-796.

ITT = intention-to-treat; RCT = randomized, controlled trial; SIRONA = Sirolimus- vs Paclitaxel-Drug Coated Balloons in Patients With Peripheral Artery Disease.

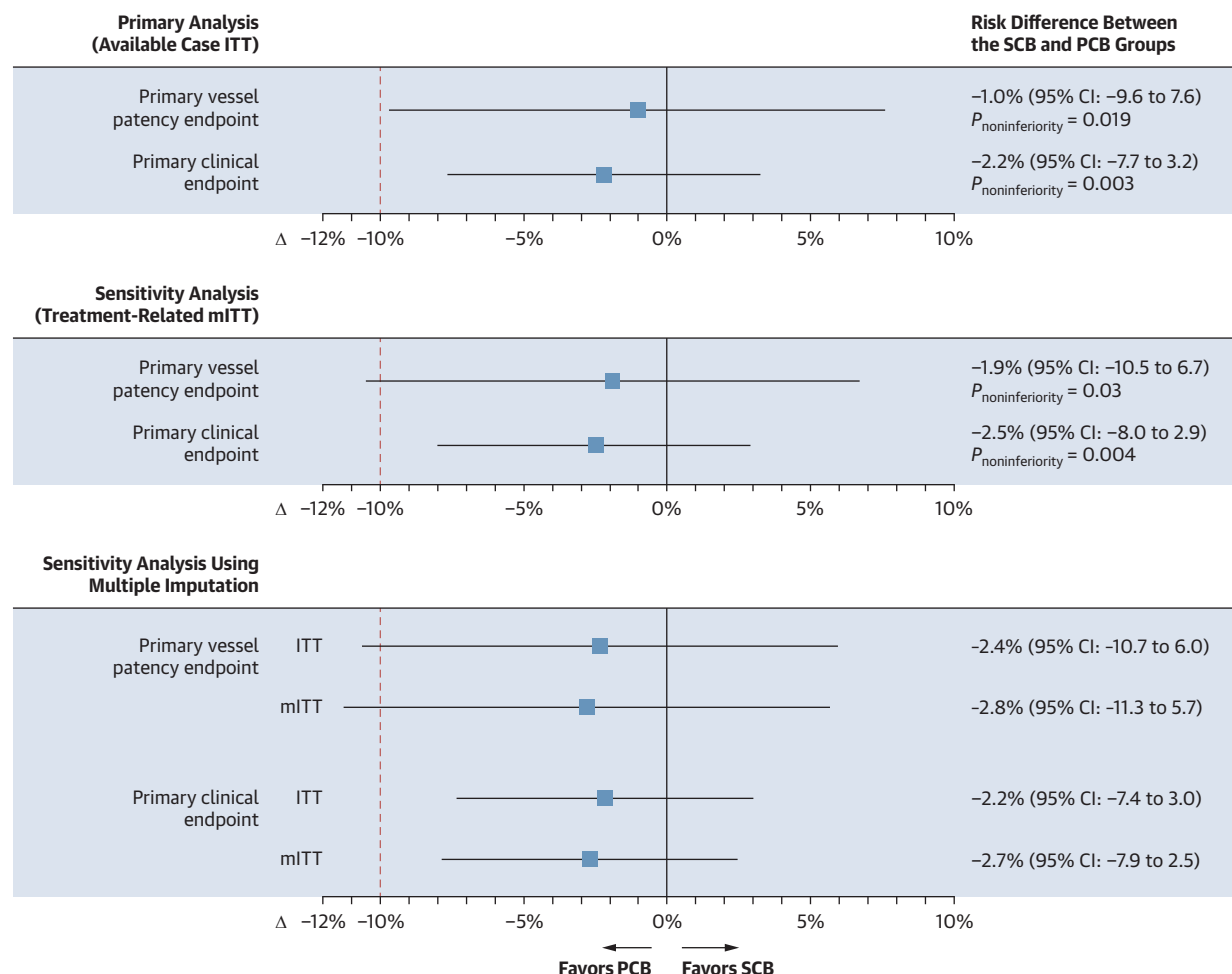
DISCUSSION

The randomized, head-to-head comparison of SCB and PCB angioplasty in the SIRONA trial showed that SCBs achieved comparable outcomes to PCBs in patients with symptomatic femoropopliteal artery disease. The improvement in the patient-related

outcomes of walking ability and vascular quality of life was similar between the 2 groups. However, sensitivity analyses that considered both actual treatment and missing data revealed that the noninferiority of 12-month primary patency was not robust.

To test noninferiority in an investigator-initiated, independent device trial of this scale and ensure

FIGURE 2 Noninferiority Test for the Primary Endpoints



Between-group comparisons were made using the Farrington-Manning test with a 1-sided alpha level of 0.025 and a noninferiority margin of 10% in favor of PCBs for each of the primary endpoints. The primary vessel patency endpoint was primary patency. The primary clinical endpoint was freedom from the composite of clinically driven target vessel revascularization, major target limb amputation, and device- or procedure-related death. The primary analysis refers to the available case ITT analysis. The treatment-related mITT analysis included only participants who received their assigned treatment. The red dashed line represents the noninferiority margin. Abbreviations as in Figure 1.

real-world feasibility, certain statistical and clinical constraints had to be accepted. Tighter noninferiority margins and a lower alpha-level would be preferable. However, the chosen thresholds reflect what could realistically be tested. The clinical constraints of a head-to-head comparison included the use of different PCB types in the control group. Therefore, this study provides an initial indication of the effectiveness of SCBs compared to PCBs, but it does not provide a generalizable positioning among DCBs.

Paclitaxel is commonly used as an anti-proliferative drug for DCBs to prevent restenosis in the femoropopliteal arteries. However, it has a

strong apoptotic effect, and high doses are required to inhibit the proliferation of hypoxic cells. Thereby, paclitaxel may increase the risk of local toxicity and downstream paclitaxel effects, such as fibrinoid necrosis.²⁹ Several cases of aneurysmal formation and vessel enlargement after subintimal PCB use have been reported.³⁰ The risk of particulate distal embolization varies among PCB types.^{29,31} Katsanos et al³² reported possible evidence of an increased risk of major amputation attributed to the cytotoxic properties of paclitaxel. Nordanstig et al¹⁰ found an increased risk of mortality within 5 years with paclitaxel-coated devices and do not support the routine use in patients with

TABLE 4 Primary and Secondary Endpoints at 12 Months

	SCB Group (n = 238)	PCB Group (n = 244)	Relative Risk (95% CI)	OR (95% CI)	RD (95% CI)* (%)	P Value (RD)
Primary endpoints	..	..	..	..	..	..
Primary patency	150/203 (73.9)	149/199 (74.9)	1.0 (0.8-1.2)	0.9 (0.6-1.5)	-1.0 (-9.6 to 7.6)	$P_{\text{noninferiority}} = 0.019^b$
Binary restenosis ^c without TLR	37/203 (18.2)	36/199 (18.1)	1.0 (0.6-1.6)	1.0 (0.6-1.7)	0.1 (-7.5 to 7.7)	>0.99
Binary restenosis followed by TLR	16/203 (7.9)	14/199 (7.0)	1.1 (0.5-2.3)	1.1 (0.5-2.4)	0.9 (-4.3 to 6.0)	0.85
Freedom from the primary clinical composite ^d	199/220 (90.5)	202/218 (92.7)	1.0 (0.9-1.1)	0.8 (0.3-1.5)	-2.2 (-7.7 to 3.2)	$P_{\text{noninferiority}} = 0.003^b$
Clinically driven TVR	21/220 (9.5)	16/218 (7.3)	1.3 (0.7-2.5)	1.3 (0.6-2.7)	2.2 (-3.1 to 7.5)	0.49
Clinically driven TLR	16/220 (7.3)	14/218 (6.4)	1.1 (0.5-2.3)	1.1 (0.5-2.4)	0.9 (-3.9 to 5.6)	0.85
Major target-limb amputation ^e	1/220 (0.5)	1/218 (0.5)	1.0 (0.1-9.5)	1.0 (0.1-9.6)	0.0 (-1.3 to 1.3)	>0.99
Device- or procedure-related death	0/220 (0)	0/218 (0)	..	..	..	..
Binary secondary endpoints	..	..	..	..	..	..
Secondary patency ^f	161/203 (79.3)	156/199 (78.4)	1.0 (0.9-1.2)	1.1 (0.6-1.8)	0.9 (-7.1 to 9.0)	0.90
Clinical improvement ^g	175/197 (88.8)	180/200 (90.0)	1.0 (0.9-1.1)	0.9 (0.4-1.7)	-1.2 (-7.3 to 4.9)	0.75
Hemodynamic improvement ^h	132/187 (70.6)	134/189 (70.9)	1.0 (0.8-1.2)	1.0 (0.6-1.6)	-0.3 (-9.6 to 8.9)	>0.99
Death from any cause ⁱ	9/231 (3.9)	5/233 (2.2)	1.8 (0.6-5.2)	1.9 (0.6-5.4)	1.7 (-1.4 to 4.9)	0.29

Values are n/N (%) unless otherwise indicated. Analysis referred to available case ITT analysis. ^aDifferences do not necessarily sum correctly due to rounding. ^bThe P value was based on the 1-sided Farrington-Manning test for noninferiority against an alpha level of 0.025 with a margin of noninferiority of 10% in favor of PCBs for each of the primary endpoints. ^cDuplex-defined binary restenosis (PSVR > 2.4) as adjudicated by the core laboratory. ^dEvent data on clinically driven TVR, major amputation, and device- or procedure-related death were included in the analysis up to 410 days after randomization. ^eMajor amputation was conducted at 7 months (SCBs) and 1 month (PCBs). It was preceded by clinically driven TLR. ^fSecondary patency was patency obtained with or without the use of an additional endovascular procedure. ^gImprovement by ≥ 1 Rutherford category. Participants who underwent TVR were excluded from the analysis. ^hImprovement in resting ankle-brachial index by ≥ 0.15 or to ≥ 0.9 . Participants with TVR were excluded from the analysis. ⁱDeath from any cause up to 410 days after the index procedure.

ITT = intention to treat; PSVR = peak systolic velocity ratio; RD = risk difference; TLR = target lesion revascularization; TVR = target vessel revascularization; other abbreviations as in Table 1.

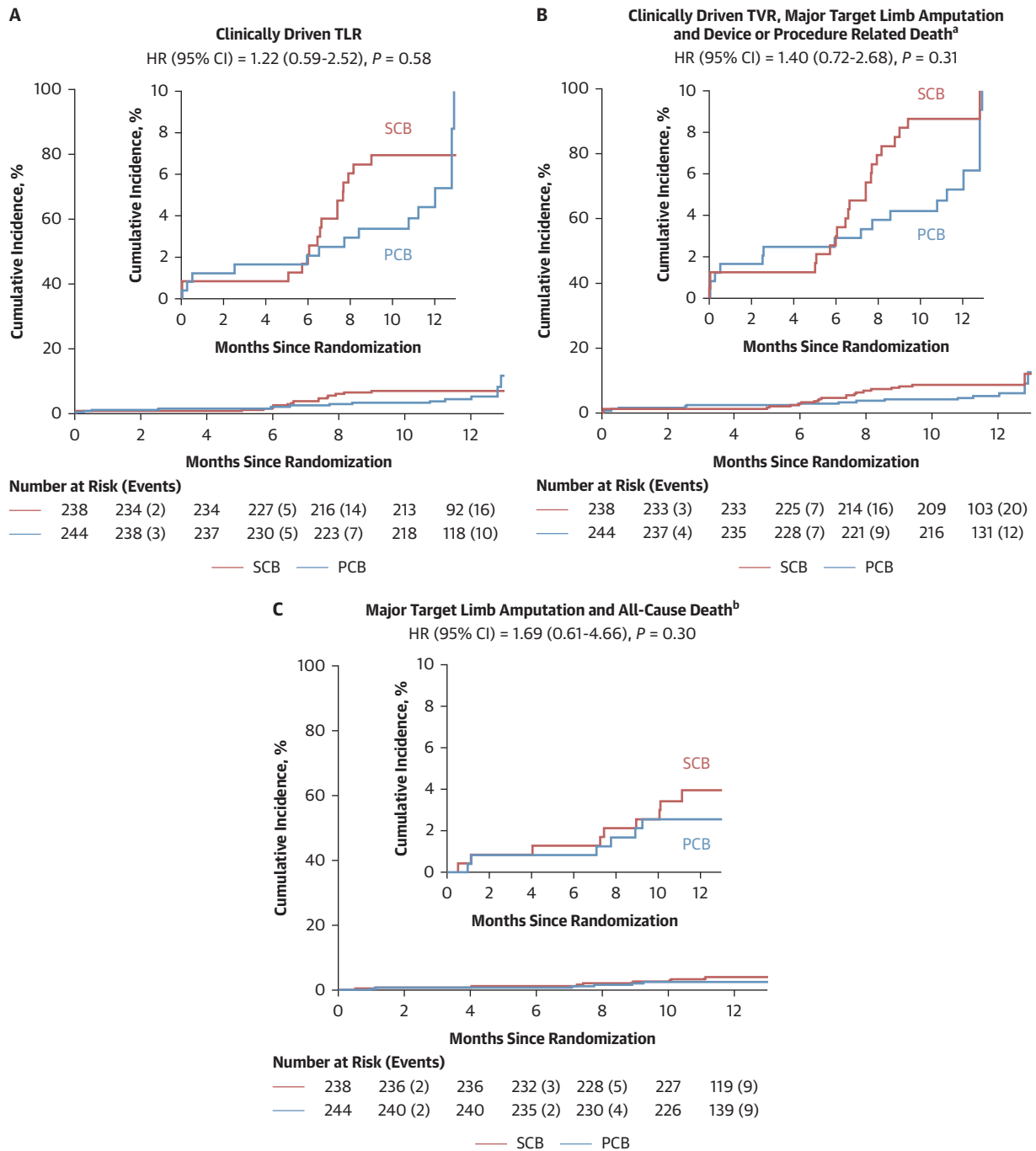
intermittent claudication. Finally, substantial heterogeneity in efficacy probably results from different coating characteristics and the paclitaxel dose density.^{5,33}

Sirolimus, on the other hand, is more effective in regulating cell proliferation in the hypoxic microenvironment of atherosclerosis and also has immunosuppressive properties.¹⁵ It has a wide therapeutic safety margin.¹⁴ As a result, paclitaxel-eluting stents have been almost completely replaced by sirolimus-eluting stents in the coronary setting. However, the sustainability of the therapeutic effect of coronary SCBs compared to sirolimus-eluting stents is currently being investigated.³⁴ After all, using the poorly lipophilic sirolimus on DCBs requires developing technologies that increase its local bioavailability and retention. Phospholipid encapsulation of sirolimus, which was used for the MagicTouch PTA (Concept Medical Inc) SCB tested in the SIRONA trial, provides nanoparticles with a lipophilic surface that enhances cellular uptake and retention. Particle size, concentration, pressure, and timing of exposure are other aspects that influence controlled release and efficient drug transfer through the layers of the vessel wall. In addition to uptake and retention, the coating's resistance to mechanical forces is essential to prevent "wash-off" with distal embolization and systemic drug leakage.²⁰

Twelve-month results of SCBs from 2 manufacturers for the treatment of femoropopliteal lesions have been reported in 3 prospective single-arm studies including a total of 234 patients.¹⁶⁻¹⁸ Primary patency was 88%, 86%, and 79% and freedom from clinically driven TLR was 97.0%, 96%, and 94%, respectively. Serious safety events were rare and mostly related to chronic limb-threatening ischemia. The treatment was therefore considered safe and efficacious. A meta-analysis showed similar results for PCB angioplasty. Twelve-month primary patency ranged from 65.2% to 86.3% in 7 RCTs (overall: 75.8%; risk ratio vs standard balloon: 1.45; $P < 0.001$) and 12-month clinically driven TLR ranged from 71% to 98% in 13 RCTs (overall: 90.3%; risk ratio vs standard balloon: 1.24; $P < 0.001$). There was substantial heterogeneity for these outcomes between different PCB types, whereas the rate of 12-month all-cause death was 2.2% without relevant heterogeneity (range 0%-4.4%; risk ratio vs standard balloon: 0.87; $P = 0.66$).⁵

The SIRONA trial protocol was designed to enroll patients with lesion morphologies reflecting clinical routine cases. Inclusion of lesions up to 20 cm in length resulted in a longer total average lesion length compared to previous pivotal DCB trials.⁵ Whereas the outcomes of the SIRONA trial were comparable to those observed in the SCB and PCB trials discussed

FIGURE 3 Clinical Events by Treatment Group



Cumulative incidences of (A) clinically driven target lesion revascularization (TLR), (B) the composite of clinically driven TVR, major target limb amputation and device or procedure related death, and (C) the composite of major target limb amputation and death from any cause. The analysis was for the ITT population. ^aAll of the events were TVRs. One TVR in each group was followed by a major amputation at 1 month (PCB group) and 7 months (SCB group), respectively. There were no deaths related to device or procedure. ^bOne of the events in each group was major amputation. Abbreviations as in Figure 1.

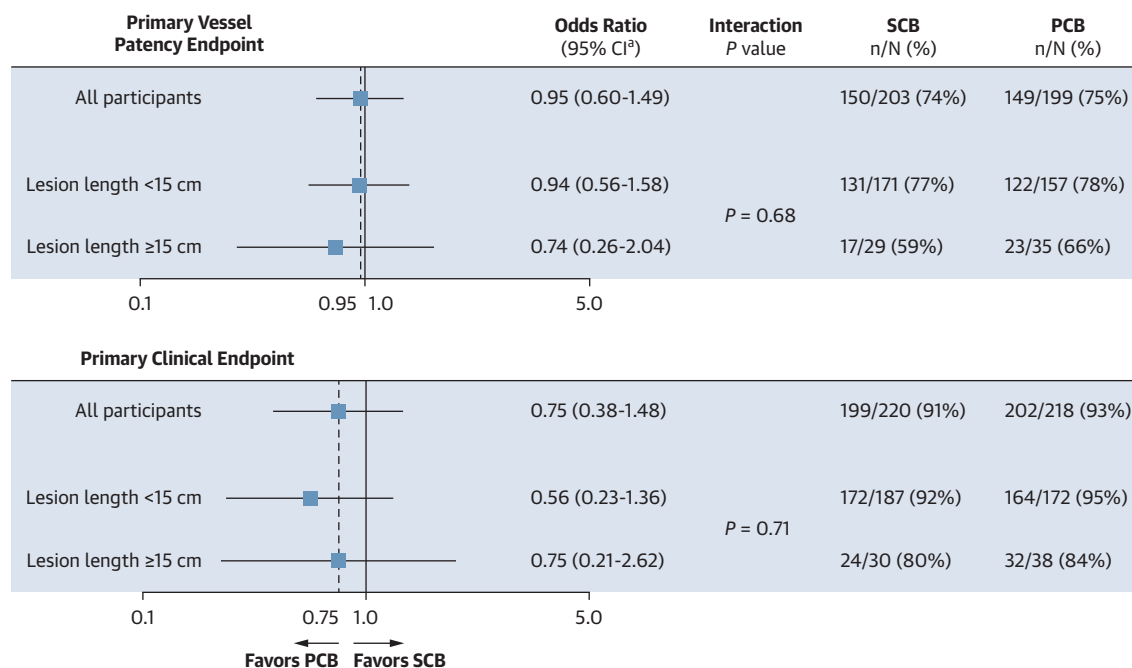
TABLE 5 Improvement in Functional Status From Baseline to 12 Months

Categorical and Continuous Secondary Endpoints ^a	SCB Group (n = 238)		PCB Group (n = 244)		Difference ^b (95% CI)	P Value
	No. Analyzed	Improvement	No. Analyzed	Improvement		
Rutherford classification ^c	197	2.0 (1.0-3.0)	200	2.0 (2.0-3.0)	OR: 0.7 (0.4-1.0)	0.043
Resting ankle-brachial index	187	0.24 ± 0.25	189	0.25 ± 0.26	−0.02 (−0.07 to 0.04)	0.48
Pain-free walking distance, m ^d	171	211 ± 174	175	190 ± 161	21 (−15 to 57)	0.25
Maximum walking distance, m ^d	172	123 ± 148	175	121 ± 144	2 (−29 to 33)	0.89
VascuQol score ^e	195	1.3 ± 1.2	197	1.4 ± 1.0	−0.03 (−0.25 to 0.20)	0.79
Pain	196	1.7 ± 1.7	203	1.6 ± 1.4	0.04 (−0.27 to 0.34)	0.82
Symptoms	196	0.8 ± 1.2	203	0.8 ± 1.1	−0.03 (−0.25 to 0.20)	0.82
Activities	197	1.7 ± 1.3	203	1.7 ± 1.2	0.01 (−0.24 to 0.26)	0.94
Social	196	1.3 ± 1.9	203	1.5 ± 1.9	−0.18 (−0.57 to 0.21)	0.35
Emotions	196	1.1 ± 1.4	202	1.1 ± 1.3	−0.06 (−0.32 to 0.21)	0.66
Health-related quality of life, EQ-5D-3L index ^f	197	0.1 ± 0.3	203	0.1 ± 0.3	−0.002 (−0.06 to 0.06)	0.94

Values are mean ± SD or median (Q1-Q3) unless otherwise indicated. Analysis referred to available case ITT analysis. ^aParticipants who underwent TVR after the index procedure were excluded from the analyses. ^bThe measure of effect is reported as the mean difference between groups or, in the case of the Rutherford category, as the OR assessed by ordinal regression. ^cThe Rutherford classification ranges from category 1 to 6. As category approaches 6, the clinical condition worsens. ^dWalking distance assessed by the 6-minute walk test. ^eThe score is derived from the 25-item VascuQol questionnaire. The domains encompass pain, symptoms, activities, social life, and emotional well-being. The score ranges from 1 to 7 (worse to best health-related quality of life). ^fThe index was calculated using the EQ-5D-3L instrument. Index ranges from 0 to 1. As the index approaches 1, the assessed quality of life increases.

EQ-5D-3L = European Quality of Life 5 Dimensions 3 Level; VascuQol = Vascular Quality of Life; other abbreviations as in Tables 1 and 4.

FIGURE 4 Subgroup Analysis by Lesion Length



Subgroups were assessed using the available case ITT analysis. The primary vessel patency endpoint was primary patency. The primary clinical endpoint was freedom from the composite of clinically driven target vessel revascularization, major target limb amputation, and device- or procedure-related death. The solid line indicates the no-effect point, and the dashed line the overall treatment effect point. ^aMiettinen-Nurminen asymptotic confidence interval. Abbreviations as in Figure 1.

herein, the SIRONA investigators appeared to underestimate the severity of lesion calcification. Notably, despite severe calcification being an angiographic exclusion criterion, the core laboratory identified such lesions (PACSS grade 4) in over one-quarter of participants. Additionally, poor distal outflow (stenosis >50%) was detected by the core laboratory in more than 10% of participants but was not reported by the investigators. Although these findings did not differ significantly between study groups, both factors may have adversely influenced outcomes in both arms.

Noninferiority of SCBs to PCBs in the SIRONA trial was achieved despite a high proportion of second-generation PCBs in the control arm, which have previously shown a comparably high efficacy (12-month primary patency: 90%, freedom from TLR: 99%).²⁷ This may be due to the paclitaxel dose density or to coating characteristics. However, the recent SATELLITE study³⁵ on long femoropopliteal lesions showed no differences in clinical outcomes according to the paclitaxel dose. As with different PCB types, there may be heterogeneity in efficacy based on sirolimus dose and coating characteristics between SCB types, which may be worth considering in future research.

In SIRONA, postdilation and bailout stenting were performed more frequently than reported in the SELUTION SFA Japan registry.¹⁸ Although, the procedural characteristics, including predilation and DCB dilation time, were comparable between the trials, the DCB inflation pressure was slightly higher in the SIRONA trial than in the registry, exceeding the nominal pressure of the MagicTouch PTA DCB on average. This could be due to severe calcification, which may have led to a “dog bone” effect with overdilation of the balloon edges and contributed to flow-limiting dissections. This assumption is also supported by the comparably high percentage of both severe calcification and bailout stenting after SCB in the all-comers registry conducted by Franzese et al.¹⁷ In general, restrained use of DCBs in severely calcified lesions and proper lesion preparation are strongly recommended for DCB angioplasty.

STUDY LIMITATIONS. In the SIRONA study the primary efficacy endpoint depended on DUS results to assess patency. However, DUS data were not available or did not meet the requirements of the core laboratory for 10% and 14% of participants in the SCB and PCB groups, respectively. In addition, data on

nontarget lesion TVR, needed to assess the primary clinical endpoint, were not available for all patients. Other reasons for incomplete data included death, withdrawal, and loss to follow-up. Therefore, primary analyses were based on the available case ITT analysis, and sensitivity analyses were performed to test the robustness of the results. In addition, the primary noninferiority endpoints were analyzed in the treatment-related mITT to adjust for participants who were assigned to the control group but erroneously received SCB angioplasty. Sensitivity analyses showed a poor robustness of noninferiority in primary patency, limiting generalizability. Another limitation was that the control group included different types of PCB, which vary in efficacy. Therefore, the results of the head-to-head comparison may vary depending on the type of PCB. Furthermore, given the relatively small proportion of long lesions, the prespecified subgroup analysis should be considered exploratory. Finally, blinding of the investigators was not feasible. Therefore, performance bias cannot be excluded.

CONCLUSIONS

The SIRONA RCT demonstrated that, in a head-to-head comparison of a representative population of patients with femoropopliteal artery disease, SCB angioplasty was noninferior to PCB angioplasty with regard to the composite clinical outcome. Additionally, similar improvements in walking ability and vascular quality of life were observed in both groups. However, the noninferiority of primary patency was narrowly achieved and was not as robust in sensitivity analyses. Therefore, SCB angioplasty may be a viable alternative, though further large-scale studies are needed to confirm its consistency and durability across devices. Long-term results of up to 5 years are awaited.

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ADDRESS FOR CORRESPONDENCE: Prof Dr. Ulf Teichgräber, Friedrich-Schiller University Jena, Jena University Hospital, Department of Radiology, Kastanienstraße 1, 07747 Jena, Germany. E-mail: ulf.teichgraeber@med.uni-jena.de.

REFERENCES

- Norgren L, Hiatt WR, Dormandy JA, et al, TASC II Working Group. Inter-society consensus for the management of peripheral arterial disease (TASC II). *J Vasc Surg*. 2007;45(suppl S):S5-S67.
- Mazzolai L, Teixido-Tura G, Lanzi S, et al, ESC Scientific Document Group. 2024 ESC guidelines for the management of peripheral arterial and aortic diseases. *Eur Heart J*. 2024;45(36):3538-3700.
- Feldman DN, Armstrong EJ, Aronow HD, et al. SCAI consensus guidelines for device selection in femoral-popliteal arterial interventions. *Catheter Cardiovasc Interv*. 2018;92(1):124-140.
- Scheinert D, Scheinert S, Sax J, et al. Prevalence and clinical impact of stent fractures after femoropopliteal stenting. *J Am Coll Cardiol*. 2005;45(2):312-315.
- Klumb C, Lehmann T, Aschenbach R, Eckardt N, Teichgräber U. Benefit and risk from paclitaxel-coated balloon angioplasty for the treatment of femoropopliteal artery disease: a systematic review and meta-analysis of randomised controlled trials. *EClinicalMedicine*. 2019;16:42-50.
- Katsanos K, Spiliopoulos S, Kitrou P, Krokidis M, Karnabatidis D. Risk of death following application of paclitaxel-coated balloons and stents in the femoropopliteal artery of the leg: a systematic review and meta-analysis of randomized controlled trials. *J Am Heart Assoc*. 2018;7(24):e011245.
- Parikh SA, Schneider PA, Mullin CM, Rogers T, Gray WA. Mortality in randomised controlled trials using paclitaxel-coated devices for femoropopliteal interventional procedures: an updated patient-level meta-analysis. *Lancet*. 2023;402(10415):1848-1856.
- Secemsky EA, Song Y, Schermerhorn M, Yeh RW. Update from the longitudinal assessment of Safety of Femoropopliteal Endovascular Treatment With Paclitaxel-Coated Devices Among Medicare Beneficiaries: the SAFE-PAD study. *Circ Cardiovasc Interv*. 2022;15(6):e012074.
- Wargny M, Leux C, Chatellier G, Coudol S, Gourraud PA, Gouëffic Y. Mortality in a nationwide practice-based cohort receiving paclitaxel-coated devices for lower limb peripheral artery disease. *J Am Coll Cardiol*. 2024;83(13):1207-1221.
- Nordanstig J, James S, Andersson M, et al, SWEDEPAD Trial Investigators. Paclitaxel-coated versus uncoated devices for infrainguinal endovascular revascularisation in patients with intermittent claudication (SWEDEPAD 2): a multicentre, participant-masked, registry-based, randomised controlled trial. *Lancet*. 2025;406(10508):1115-1127.
- Sedhom R, Hamed M, Elbadawi A, et al. Outcomes With limus- vs paclitaxel-coated balloons for percutaneous coronary intervention: meta-analysis of randomized controlled trials. *JACC Cardiovasc Interv*. 2024;17(13):1533-1543.
- Wessely R, Schömig A, Kastrati A. Sirolimus and paclitaxel on polymer-based drug-eluting stents: similar but different. *J Am Coll Cardiol*. 2006;47(4):708-714.
- Kastrati A, Dibra A, Eberle S, et al. Sirolimus-eluting stents vs paclitaxel-eluting stents in patients with coronary artery disease: meta-analysis of randomized trials. *JAMA*. 2005;294(7):819-825.
- Wessely R, Blaich B, Belaiba RS, et al. Comparative characterization of cellular and

- molecular anti-restenotic profiles of paclitaxel and sirolimus. Implications for local drug delivery. *Thromb Haemost*. 2007;97(6):1003-1012.
15. Chen Y, Zeng Y, Zhu X, et al. Significant difference between sirolimus and paclitaxel nanoparticles in anti-proliferation effect in normoxia and hypoxia: the basis of better selection of atherosclerosis treatment. *Bioact Mater*. 2021;6(3):880-889.
 16. Choke E, Tang TY, Peh E, et al. MagicTouch PTA sirolimus coated balloon for femoropopliteal and below the knee disease: results from XTOSI pilot study up to 12 months. *J Endovasc Ther*. 2022;29(5):780-789.
 17. Franzese M, Pucciarelli A, Spione F, et al. Sirolimus-coated balloon in femoropopliteal steno-occlusive disease: efficacy, safety, and 1-year outcomes. An all-comers registry. *J Endovasc Ther*. 2025;32(5):1647-1653.
 18. Iida O, Soga Y, Saito S, et al. A novel sirolimus-coated balloon for the treatment of femoropopliteal lesions: the SELUTION SFA Japan trial. *JACC Cardiovasc Interv*. 2024;17(13):1547-1556.
 19. Teichgräber U, Ingwersen M, Platzter S, et al. Head-to-head comparison of sirolimus- versus paclitaxel-coated balloon angioplasty in the femoropopliteal artery: study protocol for the randomized controlled SIRONA trial. *Trials*. 2021;22(1):665.
 20. Lemos PA, Farooq V, Takimura CK, et al. Emerging technologies: polymer-free phospholipid encapsulated sirolimus nanocarriers for the controlled release of drug from a stent-plus-balloon or a stand-alone balloon catheter. *Euro-Intervention*. 2013;9(1):148-156.
 21. Moher D, Hopewell S, Schulz KF, et al. CONSORT 2010 explanation and elaboration: updated guidelines for reporting parallel group randomised trials. *BMJ*. 2010;340:c869.
 22. Montgomery PS, Gardner AW. The clinical utility of a six-minute walk test in peripheral arterial occlusive disease patients. *J Am Geriatr Soc*. 1998;46(6):706-711.
 23. Morgan MB, Crayford T, Murrin B, Fraser SC. Developing the Vascular Quality of Life Questionnaire: a new disease-specific quality of life measure for use in lower limb ischemia. *J Vasc Surg*. 2001;33(4):679-687.
 24. Rabin R, de Charro F. EQ-5D: a measure of health status from the EuroQol Group. *Ann Med*. 2001;33(5):337-343.
 25. Steiner S, Schmidt A, Zeller T, et al. COMPARE: prospective, randomized, non-inferiority trial of high- vs. low-dose paclitaxel drug-coated balloons for femoropopliteal interventions. *Eur Heart J*. 2020;41(27):2541-2552.
 26. Scheinert D, Micari A, Brodmann M, et al, IN.PACT Global Study Investigators. Drug-coated balloon treatment for femoropopliteal artery disease. *Circ Cardiovasc Interv*. 2018;11(10):e005654.
 27. Teichgräber U, Lehmann T, Aschenbach R, et al. Efficacy and safety of a novel paclitaxel-nano-coated balloon for femoropopliteal angioplasty: one-year results of the EffPac trial. *Euro-Intervention*. 2020;15(18):e1633-e1640.
 28. Conijn AP, Jonkers W, Rouwet EV, Vahl AC, Reekers JA, Koelemay MJ. Introducing the concept of the minimally important difference to determine a clinically relevant change on patient-reported outcome measures in patients with intermittent claudication. *Cardiovasc Intervent Radiol*. 2015;38:1112-1118.
 29. Kolodgie FD, Pacheco E, Yahagi K, Mori H, Ladich E, Virmani R. Comparison of particulate embolization after femoral artery treatment with IN.PACT Admiral versus Lutonix 035 paclitaxel-coated balloons in healthy swine. *J Vasc Interv Radiol*. 2016;27(11):1676-1685.e2.
 30. Diamantopoulos A, Gupta Y, Zayed H, Katsanos K. Paclitaxel-coated balloons and aneurysm formation in peripheral vessels. *J Vasc Surg*. 2015;62(5):1320-1322.
 31. Yoshikawa M, Torii S, Aihara K, et al. Differences in biologic drug effects and distal particulate embolization in three paclitaxel-coated balloons for femoropopliteal lesions in a rabbit model. *J Endovasc Ther*. 2024;31(6):1234-1243.
 32. Katsanos K, Spiliopoulos S, Teichgräber U, et al. Editor's Choice—Risk of major amputation following application of paclitaxel coated balloons in the lower limb Arteries: A systematic review and meta-analysis of randomised controlled trials. *Eur J Vasc Endovasc Surg*. 2022;63(1):60-71.
 33. Heinrich A, Engler MS, Güttler FV, Matthäus C, Popp J, Teichgräber UK. Systematic evaluation of particle loss during handling in the percutaneous transluminal angioplasty for eight different drug-coated balloons. *Sci Rep*. 2020;10(1):17220.
 34. Yamada K, Ishikawa T, Tamura Y, Taguchi I. Superiority of coronary paclitaxel DCBs to limus DCBs in the late angiographic outcomes: the first meta-analysis showing the differences of coronary DCBs. *Cardiovasc Diagn Ther*. 2024;14(6):1025-1028.
 35. Yoshioka N, Tokuda T, Tanaka A, et al, LEADers PAD Investigators. Comparing the clinical performance of high-dose and low-dose drug-coated balloons for long femoropopliteal artery disease: results of the SATELLITE study. *Catheter Cardiovasc Interv*. 2025;105(6):1453-1461.

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APPENDIX For supplemental figures and tables; the Clinical Investigation Plan; and lists of participating committees, boards, and collaborators, please see the online version of this paper.